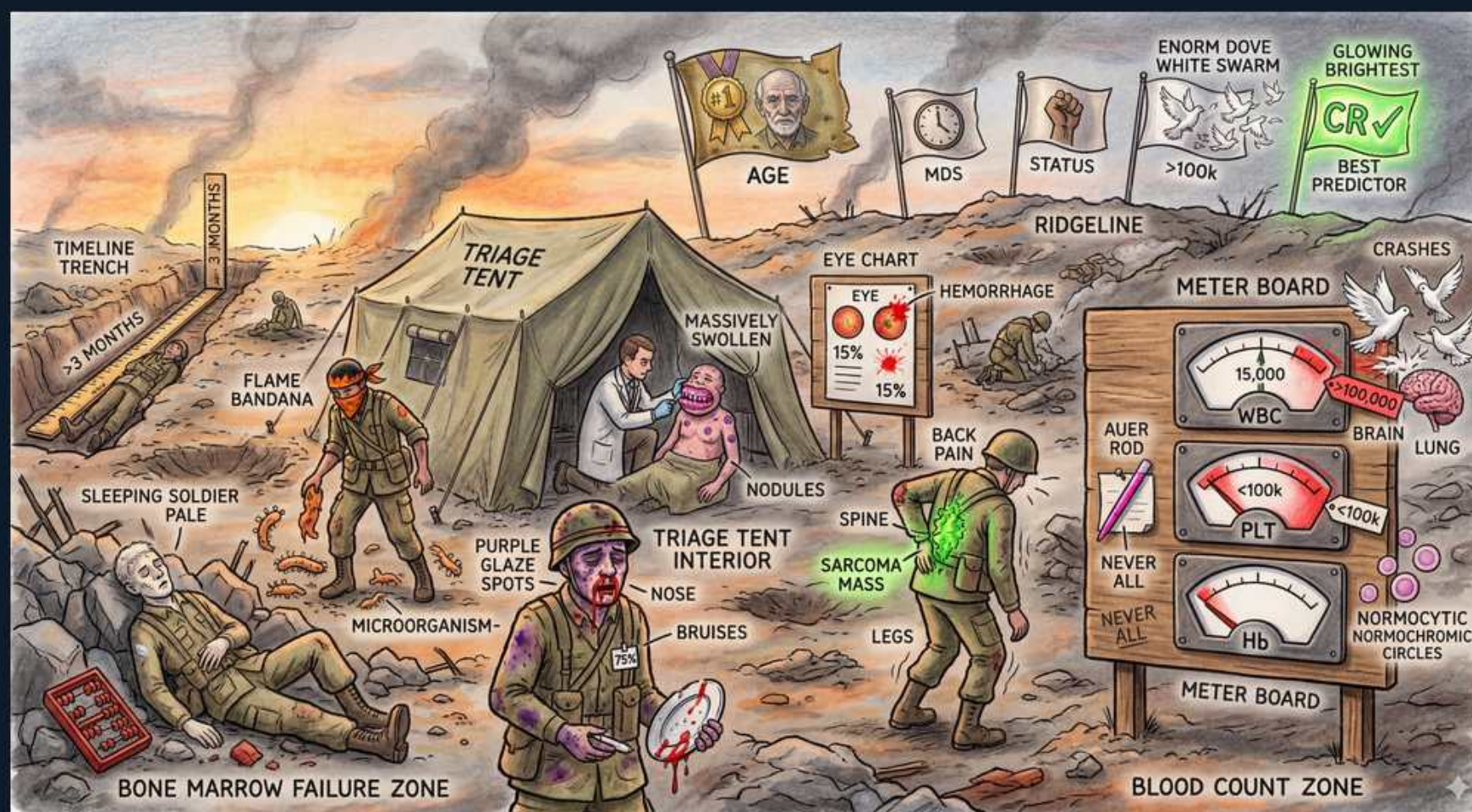


AML / Acute Leukemia — Presentation & Blood Counts



1. Older age

WHAT YOU SEE

Old portrait labeled AGE

WHAT IT ENCODES

AML is a disease of older adults, median age ~68. Incidence rises sharply with age. Older age is itself an adverse prognostic factor, partly because of higher rates of secondary AML and adverse cytogenetics, and worse tolerance of intensive induction.

BOARD TRAP

ALL is the childhood leukemia; if a vignette is an elderly adult with acute blasts, favor AML over ALL.

2. Antecedent MDS

WHAT YOU SEE

MDS clock placard

WHAT IT ENCODES

Secondary AML arising from prior MDS or MPN carries adverse risk. Look for a preceding cytopenia history. Treatment-related AML (prior alkylators/topoisomerase inhibitors) is also adverse and often has -5/-7 or MLL/KMT2A rearrangements.

BOARD TRAP

De novo AML in a young patient with t(8;21)/inv(16) is favorable — don't lump it with secondary AML's poor prognosis.

3. WHO >20% blasts

WHAT YOU SEE

STATUS sign

WHAT IT ENCODES

AML is defined by ≥20% myeloid blasts in marrow or blood. Exception: certain defining cytogenetics — t(8;21), inv(16)/t(16;16), t(15;17) — are AML at ANY blast %. Flow shows myeloperoxidase+ blasts.

BOARD TRAP

If t(15;17) or inv(16) is present with <20% blasts, still call it AML, not MDS.

4. Auer rods

WHAT YOU SEE

AUER ROD baton

WHAT IT ENCODES

Azurophilic needle-like cytoplasmic inclusions of fused primary granules — pathognomonic for myeloid lineage (AML), seen especially in APL where they form faggot cells. MPO stain positive.

BOARD TRAP

Auer rods are NEVER in ALL — their presence rules out lymphoblastic leukemia.

5. APL / bleeding (CR best predictor)

WHAT YOU SEE

Green CR check, brightest glow

WHAT IT ENCODES

Achieving complete remission (CR) after induction is the strongest predictor of long-term outcome. In APL t(15;17), ATRA + arsenic gives very high CR/cure but presents with DIC-driven bleeding requiring urgent ATRA before genetics confirm.

BOARD TRAP

Don't wait for cytogenetics in suspected APL — start ATRA immediately; delay risks fatal hemorrhage.

6. Hyperleukocytosis >100k

WHAT YOU SEE

>100k white dove swarm

WHAT IT ENCODES

WBC >100k causes leukostasis — dyspnea, hypoxia, confusion, retinal/CNS bleeds. Manage with hydroxyurea, leukapheresis, induction, and tumor lysis prophylaxis. More common in monocytic (M4/M5) subtypes.

BOARD TRAP

Avoid RBC transfusion in hyperleukocytosis — it raises viscosity and worsens leukostasis; transfuse only after cytoreduction.

7. Anemia (normocytic)

WHAT YOU SEE

Hb meter low, normocytic circles

WHAT IT ENCODES

Marrow replacement by blasts crowds out erythropoiesis → normocytic normochromic anemia with fatigue and pallor. Reticulocyte count is low (hypoproliferative).

BOARD TRAP

Macrocytosis points toward MDS or B12/folate; AML anemia is typically normocytic.

8. Thrombocytopenia

WHAT YOU SEE

PLT meter <100k red

WHAT IT ENCODES

Marrow failure drops platelets → mucocutaneous bleeding, petechiae, gum oozing. In APL, concurrent DIC compounds thrombocytopenia and bleeding.

BOARD TRAP

Schistocytes + low platelets + bleeding in a blast crisis suggests DIC (APL), not a primary microangiopathy.

9. Neutropenia / infection

WHAT YOU SEE

Triage tent, microorganism

WHAT IT ENCODES

Functional neutropenia from blast crowding causes life-threatening infection — the leading cause of early death. Febrile neutropenia mandates empiric broad-spectrum antibiotics within an hour.

BOARD TRAP

A normal total WBC can still mean profound neutropenia — count the differential, not the total.

10. Gum infiltration

WHAT YOU SEE

Massively swollen gums

WHAT IT ENCODES

Gingival hyperplasia / gum infiltration and leukemia cutis (skin nodules) are classic for monocytic AML (M4/M5, AMML/AMoL). Also associated with hyperleukocytosis.

BOARD TRAP

Gum hypertrophy = monocytic AML; it is not a feature of APL (M3).

11. Skin nodules (leukemia cutis)

WHAT YOU SEE

NODULES on torso

WHAT IT ENCODES

Leukemic skin infiltration appears as violaceous papules/nodules — again favoring monocytic subtypes and signaling extramedullary disease.

BOARD TRAP

A myeloid sarcoma (chloroma) is the solid-mass equivalent and can precede marrow involvement.

12. Myeloid sarcoma / chloroma

WHAT YOU SEE

Green SARCOMA MASS

WHAT IT ENCODES

Extramedullary tumor mass of myeloblasts (green from MPO), can occur in skin, bone, orbit, or CNS. Its presence equals AML even if marrow blasts are <20%. Often with t(8;21).

BOARD TRAP

Don't mistake a chloroma for lymphoma — flow/MPO confirms myeloid origin.

13. Bone pain / marrow failure

WHAT YOU SEE

Pale soldier, bone marrow failure zone

WHAT IT ENCODES

Marrow expansion and infiltration cause bone pain and the pancytopenia triad (anemia, infection, bleeding). This triad in an acute setting drives the marrow biopsy.

BOARD TRAP

Bone pain is more classically pediatric ALL; in adults pancytopenia + Auer rods still points to AML.

14. Symptom onset <3 months

WHAT YOU SEE

Timeline trench <3 months

WHAT IT ENCODES

Acute leukemias present over days–weeks with rapid cytopenias, contrasting the indolent months-to-years course of chronic leukemias. Rapid tempo drives urgent workup.

BOARD TRAP

A slowly rising mature lymphocytosis over years is CLL, not acute leukemia.